

URL shortner

Functional requirements:

user should be able to shorten long URL ,
all the shortened URL should be unique,
user should be to redirect to appropriate webpage once click on the shortened

Non-functional requirements:

data-partioning ,
load balancing.
fault tolerance,
vertical scalability ,
horizontal scalability,
low latency,
high throughput ,
CAP theorem (here eventual consistency is fine, we need to prioritise 99.99% uptime
i.e Availability>>Consistency)

Core Entities:

user,
url,
history,

API :
POST:shorten
{
 userID,
 url
}
GET:/click/{shortenedURL}

HLD :

